

CS5487: Machine Learning: Principles and Practice

Assignment 1 — Standard Answers

Semester A, 2026/27

Question 1

(a)

The covariance matrix is

$$\boldsymbol{\Sigma} = \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}, \quad \det(\boldsymbol{\Sigma}) = 4(1) - 1(1) = 3.$$

Therefore,

$$\boxed{\boldsymbol{\Sigma}^{-1} = \frac{1}{3} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix}}.$$

(b)

For $\mathbf{x}_A$,

$$\mathbf{x}_A - \boldsymbol{\mu} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Hence,

$$d_A^2 = (\mathbf{x}_A - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_A - \boldsymbol{\mu}) = \frac{4}{3}.$$

For $\mathbf{x}_B$,

$$\mathbf{x}_B - \boldsymbol{\mu} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

Hence,

$$d_B^2 = (\mathbf{x}_B - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_B - \boldsymbol{\mu}) = \frac{16}{3}.$$

Since both points are evaluated under the same Gaussian distribution, the point with the smaller Mahalanobis distance has the larger density. Therefore,

$$\boxed{p(\mathbf{x}_A) > p(\mathbf{x}_B)}.$$

(c)

The Gaussian normalization constant cancels in the ratio:

$$\frac{p(\mathbf{x}_A)}{p(\mathbf{x}_B)} = \frac{\exp(-d_A^2/2)}{\exp(-d_B^2/2)} = \exp \left[-\frac{1}{2} \left(\frac{4}{3} - \frac{16}{3} \right) \right].$$

Thus,

$$\boxed{\frac{p(\mathbf{x}_A)}{p(\mathbf{x}_B)} = e^2 \approx 7.389}.$$

Question 2

Write the regression model in matrix form:

$$\mathbf{X} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 2 \end{bmatrix}.$$

(a)

Under Gaussian noise, maximizing likelihood with respect to w_0 and w_1 is equivalent to minimizing the residual sum of squares.

Compute

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix}, \quad \mathbf{X}^\top \mathbf{y} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

The estimating equations are therefore

$$\begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix},$$

or

$$4w_0 + 2w_1 = 5, \quad 2w_0 + 6w_1 = 6.$$

Solving gives

$$\boxed{\hat{w}_0 = 0.9, \quad \hat{w}_1 = 0.7}.$$

(b)

The fitted values are

$$\hat{\mathbf{y}} = \mathbf{X} \hat{\mathbf{w}} = \begin{bmatrix} 0.2 \\ 0.9 \\ 1.6 \\ 2.3 \end{bmatrix}.$$

Therefore the residuals are

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} -0.2 \\ 0.1 \\ 0.4 \\ -0.3 \end{bmatrix}.$$

Hence,

$$\text{RSS} = (-0.2)^2 + (0.1)^2 + (0.4)^2 + (-0.3)^2 = 0.30.$$

For Gaussian maximum likelihood,

$$\hat{\sigma}_{\text{ML}}^2 = \frac{\text{RSS}}{n} = \frac{0.30}{4}.$$

Thus,

$$\boxed{\hat{\sigma}_{\text{ML}}^2 = 0.075}.$$

(c)

For $x_* = 3$,

$$\hat{y}_* = \hat{w}_0 + \hat{w}_1 x_* = 0.9 + 0.7(3) = 3.0.$$

Therefore,

$$\boxed{\hat{y}_* = 3.0}.$$

Question 3

The observations contain three successes and one failure. For each candidate value of π ,

$$p(\mathcal{D} \mid \pi) \propto \pi^3(1 - \pi).$$

(a)

For $\pi = 0.2$,

$$p(\mathcal{D} \mid \pi = 0.2) \propto (0.2)^3(0.8) = 0.0064.$$

Multiplying by the prior probability,

$$0.0064(0.2) = 0.00128.$$

For $\pi = 0.5$,

$$p(\mathcal{D} \mid \pi = 0.5) \propto (0.5)^3(0.5) = 0.0625,$$

so

$$0.0625(0.5) = 0.03125.$$

For $\pi = 0.8$,

$$p(\mathcal{D} \mid \pi = 0.8) \propto (0.8)^3(0.2) = 0.1024,$$

so

$$0.1024(0.3) = 0.03072.$$

The normalization constant is

$$0.00128 + 0.03125 + 0.03072 = 0.06325.$$

Therefore,

$$P(\pi = 0.2 \mid \mathcal{D}) = \frac{0.00128}{0.06325} \approx 0.0202,$$

$$P(\pi = 0.5 \mid \mathcal{D}) = \frac{0.03125}{0.06325} \approx 0.4941,$$

and

$$P(\pi = 0.8 \mid \mathcal{D}) = \frac{0.03072}{0.06325} \approx 0.4857.$$

(b)

The largest posterior probability occurs at $\pi = 0.5$. Hence,

$$\hat{\pi}_{\text{MAP}} = 0.5.$$

(c)

The posterior predictive probability averages over the full posterior:

$$P(x_* = 1 \mid \mathcal{D}) = \sum_{\pi} P(x_* = 1 \mid \pi)P(\pi \mid \mathcal{D}).$$

Since $P(x_* = 1 \mid \pi) = \pi$,

$$P(x_* = 1 \mid \mathcal{D}) = 0.2(0.0202) + 0.5(0.4941) + 0.8(0.4857).$$

Therefore,

$$P(x_* = 1 \mid \mathcal{D}) \approx 0.6396 \approx 0.640.$$

Question 4

The initial GMM parameters are

$$\pi_1^{(0)} = 0.6, \quad \pi_2^{(0)} = 0.4,$$

$$\mu_1^{(0)} = 0, \quad \mu_2^{(0)} = 4,$$

and

$$(\sigma_1^2)^{(0)} = (\sigma_2^2)^{(0)} = 2.$$

For variance 2,

$$\mathcal{N}(x \mid \mu, 2) = \frac{1}{\sqrt{4\pi}} \exp\left(-\frac{(x - \mu)^2}{4}\right).$$

The common factor $1/\sqrt{4\pi}$ cancels when responsibilities are normalized.

(a) E-step

For $x_1 = 0$,

$$s_{11} \propto 0.6e^0 = 0.6, \quad s_{12} \propto 0.4e^{-4}.$$

Thus,

$$\gamma_{11}^{(0)} = \frac{0.6}{0.6 + 0.4e^{-4}} \approx 0.9879,$$

$$\gamma_{12}^{(0)} = 1 - \gamma_{11}^{(0)} \approx 0.0121.$$

For $x_2 = 2$, both Gaussian densities are equal because the point is equally distant from the two means. Therefore,

$$\gamma_{21}^{(0)} = \frac{0.6}{0.6 + 0.4} = 0.6000,$$

$$\gamma_{22}^{(0)} = 0.4000.$$

For $x_3 = 4$,

$$s_{31} \propto 0.6e^{-4}, \quad s_{32} \propto 0.4.$$

Hence,

$$\gamma_{31}^{(0)} = \frac{0.6e^{-4}}{0.6e^{-4} + 0.4} \approx 0.0267,$$

$$\gamma_{32}^{(0)} \approx 0.9733.$$

Therefore,

	$\gamma_{i1}^{(0)}$	$\gamma_{i2}^{(0)}$
$x_1 = 0$	0.9879	0.0121
$x_2 = 2$	0.6000	0.4000
$x_3 = 4$	0.0267	0.9733

(b) Mixing weights and means

Using unrounded responsibilities,

$$N_1 = \gamma_{11}^{(0)} + \gamma_{21}^{(0)} + \gamma_{31}^{(0)} \approx 1.6147,$$

$$N_2 = \gamma_{12}^{(0)} + \gamma_{22}^{(0)} + \gamma_{32}^{(0)} \approx 1.3853.$$

Thus,

$$\boxed{\pi_1^{(1)} = \frac{N_1}{3} \approx 0.5382, \quad \pi_2^{(1)} = \frac{N_2}{3} \approx 0.4618.}$$

The new means are

$$\mu_1^{(1)} = \frac{\gamma_{11}^{(0)}(0) + \gamma_{21}^{(0)}(2) + \gamma_{31}^{(0)}(4)}{N_1} \approx 0.8094,$$

$$\mu_2^{(1)} = \frac{\gamma_{12}^{(0)}(0) + \gamma_{22}^{(0)}(2) + \gamma_{32}^{(0)}(4)}{N_2} \approx 3.3877.$$

Therefore,

$$\boxed{\mu_1^{(1)} \approx 0.8094, \quad \mu_2^{(1)} \approx 3.3877.}$$

(c) Variances

For component 1,

$$(\sigma_1^2)^{(1)} = \frac{\sum_i \gamma_{i1}^{(0)}(x_i - \mu_1^{(1)})^2}{N_1} \approx 1.0962.$$

For component 2,

$$(\sigma_2^2)^{(1)} = \frac{\sum_i \gamma_{i2}^{(0)}(x_i - \mu_2^{(1)})^2}{N_2} \approx 0.9194.$$

Hence,

$$\boxed{(\sigma_1^2)^{(1)} \approx 1.0962, \quad (\sigma_2^2)^{(1)} \approx 0.9194.}$$

Small differences in the final decimal places are acceptable if the responsibilities are rounded before the M-step.

Question 5

(a)

Since

$$Y = \mathbf{a}^\top \mathbf{X} + b,$$

linearity of expectation gives

$$\mathbb{E}[Y] = \mathbb{E}[\mathbf{a}^\top \mathbf{X} + b] = \mathbf{a}^\top \mathbb{E}[\mathbf{X}] + b.$$

Therefore,

$$\boxed{\mathbb{E}[Y] = \mathbf{a}^\top \boldsymbol{\mu} + b}.$$

(b)

Using part (a),

$$Y - \mathbb{E}[Y] = \mathbf{a}^\top \mathbf{X} + b - (\mathbf{a}^\top \boldsymbol{\mu} + b) = \mathbf{a}^\top (\mathbf{X} - \boldsymbol{\mu}).$$

Therefore,

$$\text{Var}(Y) = \mathbb{E} \left[\left(\mathbf{a}^\top (\mathbf{X} - \boldsymbol{\mu}) \right)^2 \right].$$

Because this is scalar,

$$\left(\mathbf{a}^\top (\mathbf{X} - \boldsymbol{\mu}) \right)^2 = \mathbf{a}^\top (\mathbf{X} - \boldsymbol{\mu}) (\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{a}.$$

Hence,

$$\text{Var}(Y) = \mathbf{a}^\top \mathbb{E} \left[(\mathbf{X} - \boldsymbol{\mu}) (\mathbf{X} - \boldsymbol{\mu})^\top \right] \mathbf{a}.$$

Using

$$\boldsymbol{\Sigma} = \mathbb{E} \left[(\mathbf{X} - \boldsymbol{\mu}) (\mathbf{X} - \boldsymbol{\mu})^\top \right],$$

we obtain

$$\boxed{\text{Var}(Y) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}}.$$

Question 6

The observations are independent and

$$x_i \sim \mathcal{N}(\mu, \sigma_i^2),$$

where the variances σ_i^2 are known.

(a)

The likelihood is

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp \left[-\frac{(x_i - \mu)^2}{2\sigma_i^2} \right].$$

Taking logarithms,

$$\ell(\mu) = -\frac{1}{2} \sum_{i=1}^n \log(2\pi\sigma_i^2) - \frac{1}{2} \sum_{i=1}^n \frac{(x_i - \mu)^2}{\sigma_i^2}.$$

Ignoring terms independent of μ ,

$$\boxed{\ell(\mu) = -\frac{1}{2} \sum_{i=1}^n \frac{(x_i - \mu)^2}{\sigma_i^2} + \text{constant} .}$$

(b)

Differentiate:

$$\frac{d\ell}{d\mu} = \sum_{i=1}^n \frac{x_i - \mu}{\sigma_i^2}.$$

At the maximum,

$$\sum_{i=1}^n \frac{x_i - \hat{\mu}}{\sigma_i^2} = 0.$$

Thus,

$$\sum_{i=1}^n \frac{x_i}{\sigma_i^2} = \hat{\mu} \sum_{i=1}^n \frac{1}{\sigma_i^2}.$$

Therefore,

$$\boxed{\hat{\mu}_{\text{ML}} = \frac{\sum_{i=1}^n x_i / \sigma_i^2}{\sum_{i=1}^n 1 / \sigma_i^2} .}$$

(c)

If

$$\sigma_1^2 = \dots = \sigma_n^2 = \sigma^2,$$

then

$$\hat{\mu}_{\text{ML}} = \frac{(1/\sigma^2) \sum_i x_i}{n/\sigma^2} = \frac{1}{n} \sum_i x_i.$$

Hence,

$$\boxed{\hat{\mu}_{\text{ML}} = \bar{x} .}$$

Question 7

The likelihood is

$$p(\mathcal{D} \mid w) \propto \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - w)^2 \right],$$

and the prior is

$$p(w) \propto e^{-\lambda|w|}.$$

(a)

The negative log-posterior, ignoring constants independent of w , is

$$J(w) = \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - w)^2 + \lambda|w|.$$

Using

$$\sum_{i=1}^n (x_i - w)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(w - \bar{x})^2,$$

the first term on the right that does not depend on w can be dropped. Thus,

$$J(w) = \frac{n}{2\sigma^2} (w - \bar{x})^2 + \lambda|w| + \text{constant}.$$

(b)

For $w > 0$,

$$|w| = w,$$

so

$$\frac{dJ}{dw} = \frac{n}{\sigma^2} (w - \bar{x}) + \lambda.$$

Setting the derivative to zero,

$$w = \bar{x} - \frac{\lambda\sigma^2}{n}.$$

This is valid for $w > 0$ when

$$\bar{x} > \frac{\lambda\sigma^2}{n}.$$

For $w < 0$,

$$|w| = -w,$$

so

$$\frac{dJ}{dw} = \frac{n}{\sigma^2} (w - \bar{x}) - \lambda.$$

Setting the derivative to zero,

$$w = \bar{x} + \frac{\lambda\sigma^2}{n}.$$

This is valid for $w < 0$ when

$$\bar{x} < -\frac{\lambda\sigma^2}{n}.$$

(c)

When

$$|\bar{x}| \leq \frac{\lambda\sigma^2}{n},$$

the optimum occurs at $w = 0$.

Therefore,

$$\hat{w}_{\text{MAP}} = \text{sign}(\bar{x}) \max \left(|\bar{x}| - \frac{\lambda \sigma^2}{n}, 0 \right).$$

Equivalently,

$$\hat{w}_{\text{MAP}} = \begin{cases} \bar{x} - \frac{\lambda \sigma^2}{n}, & \bar{x} > \frac{\lambda \sigma^2}{n}, \\ 0, & |\bar{x}| \leq \frac{\lambda \sigma^2}{n}, \\ \bar{x} + \frac{\lambda \sigma^2}{n}, & \bar{x} < -\frac{\lambda \sigma^2}{n}. \end{cases}$$

Question 8

For every component,

$$\pi_k = \frac{1}{K}, \quad \Sigma_k = \sigma^2 \mathbf{I}.$$

(a)

By Bayes' rule,

$$\gamma_{ik} = P(z_i = k \mid \mathbf{x}_i) = \frac{\pi_k \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \sigma^2 \mathbf{I})}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_j, \sigma^2 \mathbf{I})}.$$

For the isotropic Gaussian,

$$\mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \sigma^2 \mathbf{I}) = C \exp \left[-\frac{\|\mathbf{x}_i - \boldsymbol{\mu}_k\|_2^2}{2\sigma^2} \right],$$

where C is the same for every component.

Since the mixing weights are also equal, both C and $1/K$ cancel. Hence,

$$\gamma_{ik} = \frac{\exp \left[-\frac{\|\mathbf{x}_i - \boldsymbol{\mu}_k\|_2^2}{2\sigma^2} \right]}{\sum_j \exp \left[-\frac{\|\mathbf{x}_i - \boldsymbol{\mu}_j\|_2^2}{2\sigma^2} \right]}.$$

(b)

For components a and b ,

$$\frac{\gamma_{ia}}{\gamma_{ib}} = \frac{\exp \left[-\frac{\|\mathbf{x}_i - \boldsymbol{\mu}_a\|_2^2}{2\sigma^2} \right]}{\exp \left[-\frac{\|\mathbf{x}_i - \boldsymbol{\mu}_b\|_2^2}{2\sigma^2} \right]}.$$

Therefore,

$$\frac{\gamma_{ia}}{\gamma_{ib}} = \exp \left[\frac{\|\mathbf{x}_i - \boldsymbol{\mu}_b\|_2^2 - \|\mathbf{x}_i - \boldsymbol{\mu}_a\|_2^2}{2\sigma^2} \right].$$

(c)

Let k^* be the unique nearest centroid. Then for every $j \neq k^*$,

$$\|\mathbf{x}_i - \boldsymbol{\mu}_j\|_2^2 - \|\mathbf{x}_i - \boldsymbol{\mu}_{k^*}\|_2^2 > 0.$$

Factor the k^* term out of the responsibility denominator:

$$\gamma_{ik^*} = \frac{1}{1 + \sum_{j \neq k^*} \exp \left[-\frac{\|\mathbf{x}_i - \boldsymbol{\mu}_j\|_2^2 - \|\mathbf{x}_i - \boldsymbol{\mu}_{k^*}\|_2^2}{2\sigma^2} \right]}.$$

As $\sigma^2 \rightarrow 0$, every exponential term in the sum tends to zero. Hence,

$$\gamma_{ik^*} \rightarrow 1$$

and, for every $j \neq k^*$,

$$\gamma_{ij} \rightarrow 0.$$

Thus the soft GMM assignments become hard nearest-centroid assignments in the small-variance limit.

(d)

For fixed hard assignments,

$$J_k(\boldsymbol{\mu}_k) = \sum_i r_{ik} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|_2^2.$$

Differentiate:

$$\nabla_{\boldsymbol{\mu}_k} J_k = 2 \sum_i r_{ik} (\boldsymbol{\mu}_k - \mathbf{x}_i).$$

At the minimum,

$$\sum_i r_{ik} (\boldsymbol{\mu}_k - \mathbf{x}_i) = 0.$$

Therefore,

$$\boldsymbol{\mu}_k \sum_i r_{ik} = \sum_i r_{ik} \mathbf{x}_i.$$

Hence,

$$\boxed{\boldsymbol{\mu}_k = \frac{\sum_i r_{ik} \mathbf{x}_i}{\sum_i r_{ik}}}.$$